Automatic SAR Ship Detection Based on Multifeature Fusion Network in Spatial and Frequency Domains

Shiyu Wang, Zhanchuan Cai, *Senior Member, IEEE*, and Jieyu Yuan

***Abstract*— Synthetic aperture radar (SAR) ship detection is sensitive to the interference of inshore background, disturbance of strong wind and waves. The similar textures of the neighbor objects in SAR images affect the detection performance. As a remarkable indicator, textural information in the frequency domain characterizes the subtle textural differences between an object and its surroundings. Inspired by this, a multifeature fusion network (MFFN) for SAR ship detection is constructed in this article, which can obtain contour and detail information of an SAR image for detecting ships from their background. First, spatial and frequency information of ship targets, which characterizes the whole and subtle textural information of ship targets, are extracted by a double-backbone network with Haar wavelet transform. Afterward, a binary domain feature pyramid network (BDFPN) with feature fusion block (FFB) is applied to fuse the spatial, frequency textural information of ship targets to obtain the fused feature maps with a top-down structure. Finally, those feature maps are adopted through the region proposal network (RPN) for detecting ship targets from original images. The experimental results show that the proposed method achieves greater performance and more accurate detection results in unique situations in the state-of-the-art SAR ship detection dataset (SSDD).**

***Index Terms*— Fusion features, Haar wavelet transform, ship detection, spatial and frequency domain, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave sensor that can obtain high-resolution data and is widely used in early coastal warning and maritime supervision fields, such as ocean monitoring, ocean management and Earth observation and so on [1], [2], [3], [4], [5]. For ship detection, SAR image has the advantages of high-precision imaging and high-precision detection and has high detection and recognition ability for small changes of targets. Compared with traditional passive imaging sensors, SAR is rarely affected by environmental factors like light and clouds during the imaging process, so it can generate high-resolution microwave images under different weather conditions and several observation scenes. Especially with the launch and operation of a new generation of spaceborne high-resolution SAR satellites, the potential of ship recognition in SAR images has been further enhanced. So far, SAR images are still the common database for ship detections.

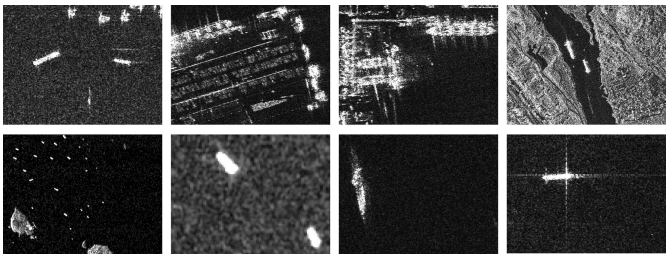


Fig. 1. Typical SAR images from the SSDD dataset in inshore and offshore scenes, including different types of scenes such as ports, islands and reefs, and sea surfaces with different levels of sea conditions.

Typical situations of SAR ship distributions are shown in Fig. 1, containing offshore and inshore scenes. There are also different types of complex backgrounds, including ports, islands, dams or other objects similar to ships, as well as different sea surface conditions caused by ship wakes and waves, which makes detection more difficult to distinguish ships from other objects. At the same time, SAR images have complex scale variations, which also become one of the main factors interfering with the accuracy of ship detection.

So far, numerous methods have been proposed to detect SAR ship targets. The traditional SAR ship detection method extracts simple low-level features and requires a lot of manpower. With the rise of artificial intelligence and the rapid development of SAR imaging technology, convolutional neural network (CNN) based methods have achieved remarkable success in computer vision tasks, and some accurate and robust ship target detection methods based on CNN with excellent detection performance in high-resolution SAR images have also emerged. Using the refine semantic information of the target, the detection accuracy of the SAR ship target can be further improved, and it is not as complex as traditional algorithms.

These existing SAR ship detection methods are verified that show manual and time-consuming performance, also paying more attention to salient objects from a single and salient

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feature information (i.e., large-scale ships in the offshore scene in SAR images). In practical applications, due to the inference of sea surface conditions and complex background, it is difficult to distinguish ship wakes, other objects, and ship targets in inshore scene. Therefore, ships closer to the coast are easily ignored, which makes the performance of SAR ship detection limited. Few of them are suitable for detecting multiscale ships in inshore scene.

To overcome the inference of sea waves and complex background, we propose a multifeature fusion network (MFFN) for SAR ship detection in this article, which can extract frequency domain features to make full use of different levels of features to combine spatial, high-frequency, and low-frequency textural information for reducing the interference of multiscale and complex background in ship detection. First, a double backbone network with Haar wavelet transform is used to obtain the spatial and frequency information of the ship targets. The original image can be divided into four sub-images through the 2-D Haar wavelet transform. These sub-images contain the information in the horizontal, vertical, and diagonal directions. By combining the 2-D Haar wavelet transform and the CNN, we can obtain the spatial and frequency features, which can improve the accuracy of the network to distinguish between inshore objects and ship targets. Next, a binary domain feature pyramid network (BDFPN) with feature fusion block (FFB) is used to extract the spatial, contour, and detail features of the ship targets with a top-down structure. FFB fuses the spatial and frequency features, which characterizes the whole and subtle textural information of ship targets, enhances the relationship between local features and their global relevance, and highlights important information. Finally, the fused features will be adopted to detect ship targets. Using the SAR ship detection dataset (SSDD) [6], the effectiveness and feasibility of the proposed method are verified, and the experiments show that the proposed method can more accurately distinguish ship targets from other objects, especially in inshore backgrounds. Because of the use of two domain features, more accurate detection results are obtained. The major contributions of this article are as follows.

1) *MFFN:* is proposed to detect multiscale ships for SAR images. The background interference is alleviated through the fused feature, and the accuracy of ship detection is increased. The proposed method can more accurately distinguish ship targets from other targets, especially in inshore scene.
2) *Textural Information of Neighbor Objects:* is refined by the Haar wavelet transform. The de-noised textural feature can characterize sample differences to enhance the detection performance.
3) *BDFPN:* is designed, which fuses the spatial, high-frequency, and low-frequency features extracted from the spatial and frequency domains, creating the multiscale fused features of the ship targets with a top-down pyramid structure.

The rest of this article is as follows. Section II introduces the related work of SAR ship detection. Section III describes the detailed procedures of the proposed methodology. Section IV shows the experimental results and analysis of the method using SSDD. Section V is the discussion part. Finally, the fifth section is a summary of our work.

II. Related Work

In this section, we introduce some traditional SAR ship detection algorithms and some CNN-based SAR ship detection algorithms.

Based on backscatter, polarization, and geometric characteristics, these traditional detection methods can be roughly divided into three categories. The constant false alarm rate (CFAR) detection algorithm [7] and its extended versions [8], [9], [10] take advantage of the characteristics of backscattering and have been widely used in SAR ship detection. The CFAR method adaptively adjusts the detection threshold according to the established statistical distribution model of background clutter, and the most appropriate threshold can be obtained for the ship target. However, the development of accurate models, such as the statistical distribution of background clutter is vulnerable to wave and environmental changes. Therefore, it also brings great difficulties and challenges to detect the CFAR of ship targets near the coast, ports, and reefs using an accurate statistical model. The central idea of the polarization-based method is to distinguish between targets and background using the difference of scattering mechanism between ship targets and sea clutter [11], [12]. In recent years, with the launch of the PolSAR satellite, the detection performance of PolSAR has been significantly improved due to the abundant scattering information compared with that of single-polarization SAR images [13], [14], [15]. However, it is not easy to establish a comprehensive library of polarization scattering characteristics. In addition, due to the high requirements for the detection environment, algorithms based on polarization characteristics are also vulnerable to the influence of the sea clutter background. Using geometric features such as shape, area, size, and texture [16], [17], a template matching strategy is also a way of ship detection. Since these features directly reflect the physical characteristics of SAR ship targets and they have become the dominant SAR ship target detection methods [18]. However, the target detection performance of those methods largely depends on the template library established using expert experiences. It is difficult to apply them to other ship detection scenes because of the need to match all SAR image pixels with the template and their transmission capacity is not satisfactory, which increases the computing cost.

CNN-based methods can be divided into two categories according to the detector, one-stage detector and two-stage detector. One-stage detectors, such as single-shot multibox detector (SSD) [19], you only look once (YOLO) [20], [21], and RetinaNet [22], can directly predict locations and class labels of potential objects at several feature maps without cropping and resizing region of interest (RoI) proposals. In [23], to achieve low cost, a simplified U-Net [24] SAR ship detection is proposed, but the detection performance is unsatisfactory. In [25], the DarkNet-53 is adopted to combine multiscale features into a SSD network to improve multiscale ship detection performance. In two-stage detectors,

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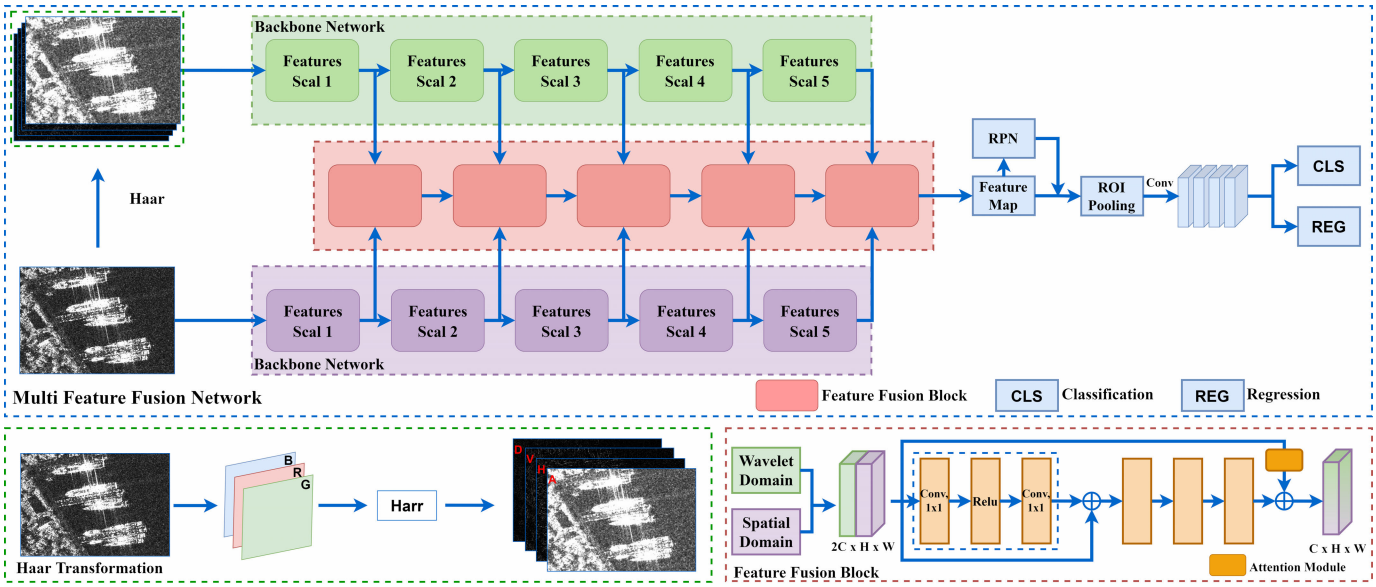


Fig. 2. Processing scheme of our method. First, a double backbone network with Haar wavelet transform is used to obtain the textural and semantic information of the target. Next, a BDFPN with a FFB is used to extract the spatial, contour, and detail features of the ship targets with a top-down structure. Finally, the fused features will be adopted to detect ship targets.

basic features are extracted by backbone networks, such as VGG [26], ResNet [27], Inception [28], and DenseNet [29], and RoI proposals are generated by region proposal network (RPN) [30]. Features of these proposals are resized to fixed sizes and processed by two branches of classification and regression networks.

By taking advantage of the low-level texture and high-level semantic features, a modified Faster RCNN method called as multilayer fusion light-head detector (MFLHD) is developed to improve the ship target detection performances [31]. In [32], a squeeze and excitation rank Faster RCNN (SER Faster-RCNN) method is proposed, and compared with the existing RCNN method, it obtains a better detection performance by learning the target's multiscale characteristics and introducing the squeeze and excitation mechanism. In [33], the basics structure of FPN [34] adopted a convolutional block attention module (CBAM) [35] at a top-down pathway of the lateral connections called dense attention pyramid network (DAPN), which can obtain more refined multiscale feature expressions and filter out negative objects and suppress interference of surroundings. A novel method named attention receptive pyramid network (ARPN) is proposed [36], RFB and CBAM are utilized to improve the performance of detecting multiscale ships in complex SAR scenes. A multidimensional domain deep learning network (PolFourier) for SAR ship detection is proposed [37], where the frequency-domain features that are not sensitive to ship target rotation are obtained by the polar Fourier transform [38], making use of complementary features in space and frequency domain performs well in complex background. The one-stage detector sacrifices the processing accuracy because of its simple processing scheme, feature extraction and recognition strategy. Although the processing speed of two-stage detectors is reduced due to a large number of RoI suggestions and complex processing schemes, their well-designed feature representation modules can achieve high-detection accuracy.

To sum up, the traditional SAR ship detection method only extracts simple low-level features, and requires a lot of manpower. These CNN-based SAR ship detection methods only use a single and salient feature information to detect the targets, which is easy to be affected by sea clutter and complex background and is difficult to distinguish ship wakes, other objects, and ship targets in inshore scene; ships near the coast are easy to be ignored. Therefore, we need an algorithm that does not limit the performance of SAR ship detection and is suitable for detecting multiscale ships in inshore scene.

III. Methodology

In this section, the backbone feature fusion network is developed, and details of the implementation procedures are also presented. First, the overall structure of the network is obtained based on the defects of existing methods. After that, every block of the proposed methods is described in detail to show how it works. Finally, in this section, the loss function of the proposed method is given.

A. Overview

In this article, we construct a MFFN for SAR ship detection, which is a two-stage detector. MFFN can transform the original image information into the frequency domain, and extract the features of the spatial domain and frequency domain. Then it fuses those features to make up for the lack of textural features in spatial domain features. Thus, MFFN can reduce the interference of inshore background, disturbance of strong wind and waves, and the effect with the similar textures of the neighbor objects, and make the detection more efficient and accurate. The processing scheme of our method is shown in Fig. 2.

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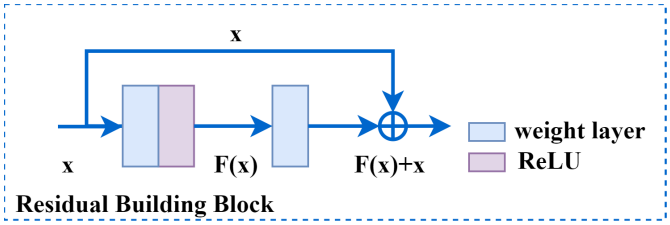


Fig. 3. Architecture of residual building block. The mapping from the input feature x to the residual $F(x)$ is completed by a network block composed of two weight layers, and the output of the network block is $F(x)+x$.

In the first stage, Haar wavelet transform is used to transform original information into the frequency domain. And ResNet as a backbone in MFFN extracts features of ship targets in the spatial domain and frequency domain. The multiscale fused features of ship targets are obtained with a top-down structure by BDFPN with FFB. Then we utilize the RPN to generate a series of the candidate regions with spatial-frequency features. And the location information of the candidate region can be mapped into the feature maps and original image to obtain the corresponding ship targets image. In the second stage, an RoI warping layer [39] is used to crop and resize RoIs into a fixed size from feature maps. Then, these RoIs are sent to classification and regression subnetworks for discrimination and localization tasks, respectively. The final detection results are acquired by merging and discarding redundant potential objects.

B. Multiscale Spatial and Frequency Feature Extraction

In this article, we construct a double backbone network to obtain spatial and frequency domain features from SAR images with ResNet [27]. Textural information in the frequency domain is verified as a remarkable indicator for characterizing the subtle textural differences between an object and its surroundings. Therefore, one backbone abstracts semantic information through ResNet, and another one adds a Haar wavelet to transform spatial features into frequency features to get more textural information, which can decrease the interference of inshore background, disturbance of strong wind and waves, and the effect with the similar textures of the neighbor objects.

1) Deep Residual Network: Residual network (ResNet), as a typical feature extraction network, is widely used in ship object detection in SAR images [33], [36], [37]. Therefore, we apply ResNet to set up our double backbone network. ResNet is built by a residual building block, which is composed of multiple concatenated convolution layers and a shortcut connection, as shown in Fig. 3.

This network block maps the input feature x to the output feature $H(x)$, records $F(x)$ as the difference between the original mapping $H(x)$ and the input feature x, and the original mapping is recast into $F(x)+x$.

The mapping from the input feature x to the residual $F(x)$ is completed by a network block composed of two weight layers, which represents refers to the convolution layer and batch normalization layer. The operation $F(x)+x$ is performed by a shortcut connection and element-wise addition. The structure in Fig. 3 can be expressed as the following equation:

$$y = F(x, \{W_i\}) + x \tag{1}$$

where $F = W_2\sigma(W_1 x)$ and σ represents a ReLU activation function [40]. Then the element-wise addition of $F(x)$ and x is completed and $\sigma(y)$ is the final output of the structure in Fig. 3. The shortcut connections in Eq. (1) will not introduce extra parameters or computation complexity.

ResNet retains the original features while extracting the spatial features of the image through convolution. Thus, we use ResNet as the backbone to extract the basic features in the spatial domain and frequency domain, which can obtain more detailed features.

2) Haar Wavelet Transform: Haar wavelet transform is a nonlinear wavelet transform method [41], which has the multiresolution features of the wavelet decomposition. These features make Haar wavelet have outstanding performance in information extraction.

The basic idea of the wavelet analysis is to use a family of functions to express or approximate signals [42]. This family of functions is called a wavelet function system, which is formed through the expansion and translation of a basic wavelet function at different scales. And Haar wavelet transform can be expressed in the following matrix form:

$$T = HFH^T \tag{2}$$

where F is the image matrix, H is the Haar transform matrix (H is asymmetric) and contains the Haar basis function $h_{(z)}$. They are defined on continuous closed interval $z \in [0, 1]$, $k = 0, 1, 2, \ldots, N-1$, where $N = 2^n$. To generate the matrix H, define the integer $k = 2^p + q - 1$, where $0 \le p \le n-1$, when $p = 0$, $q = 0$, or 1, when $p \ne 0$, $1 \le q \le 2^p$. Therefore, the Haar basis function can be expressed as follows:

$$h_0 z = h_{00} z = \frac{1}{\sqrt{N}}, \quad z \in [0, 1] \tag{3}$$

$$h_k(z) = h_{pq}(z) = \frac{1}{\sqrt{N}} \begin{cases} 2^{\frac{p}{2}}, & \frac{q-1}{2^p} \le z < \frac{q-0.5}{2^p} \\ -2^{\frac{p}{2}}, & \frac{q-0.5}{2^p} \le z < \frac{q}{2^p} \\ 0, & z \in [0, 1] \end{cases}. \tag{4}$$

The wavelet transform of images is the basis for the application of wavelets to image analysis and feature extraction. These wavelets measure intensity variation for images along different directions. The decomposition of the image by a 2-D discrete wavelet transform can be seen as the filtering process shown in Fig. 4.

At the beginning of the Haar wavelet transform, convolving the rows of the input SAR image and downsampling their columns, we get two sub-images whose horizontal resolutions are reduced by a factor of 2. The image's high-frequency information with vertical orientation is characterized by the high-pass or detail component, and the low-pass, approximation component contains its low-frequency vertical information. In the low-frequency sub-image, the edges and details of the SAR image are well preserved, while the noise is distributed in the high-frequency sub-image. Sub-images are then filtered columnwise and downsampled to yield four quarter-size output

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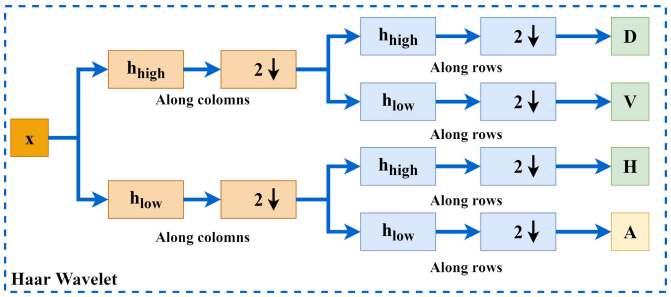


Fig. 4. Haar wavelet transform. First, do 1-D row filter along each column of the matrix to obtain high-frequency and low-frequency components and go through a down sampling, then filter the results along the row direction, and do a down sampling again. After all the column filtering, four different frequency bands are obtained, including one approximate component and three detail components.

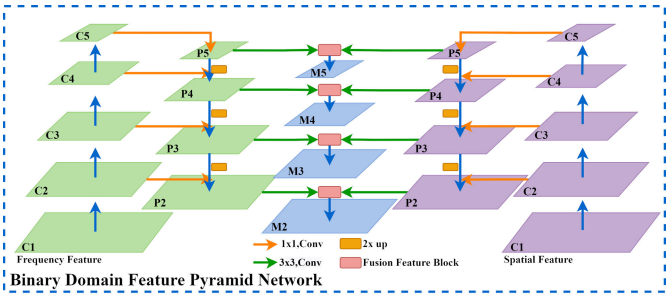


Fig. 5. BDFPN includes bottom-up feed-forward networks, FFB, transverse connections and top-down up-sampling processes. BDFPN fuses the spatial features extracted from the spatial domain and high-frequency and low-frequency textural information extracted by Haar wavelet transform, which can better extract the multiscale features of the ship target in two domains.

sub-images: W_A is the approximation coefficient after low-pass filtering in the horizontal and vertical directions, which is the most important coefficient and contains the approximation information of the original image. At the same time, random noise and redundant information are eliminated in these components. W_V is the vertical detail coefficient after low-pass filtering in the horizontal direction and high-pass filtering in the vertical direction, including the characteristics in the horizontal direction. W_H is the horizontal detail coefficient in the vertical direction, where the edges and textures of the ship correspond to the strong response. W_D is the diagonal detail coefficient after high-pass filtering in the horizontal and vertical directions, which contains the least information.

After the first-order decomposition operations, the Gaussian noise and salt-and-pepper noise in the horizontal high-frequency sub-images and the vertical high-frequency sub-images can be filtered by high-pass filtering to achieve the purpose of noise removal. SAR feature information will be clearer, which effectively improves the efficiency and accuracy of the algorithm in the information extraction process. Therefore, this article selects four sub-images as the input features in the frequency domain. The values of most points on the high-resolution sub-images are close to 0. The higher the resolution, this phenomenon becomes more obvious. The higher the decomposition level of the sub graph, the lower the frequency will be. Therefore, we only use the sub graph after the first-level wavelet transform to extract the contour and detail features in the frequency domain.

C. Binary Domain Feature Pyramid Network

In convolutional networks, deep networks can obtain semantic features, while shallow networks can extract image features. However, in object detection, the high-level network, due to the small size of the feature map, it has little location information. Thus, the high-level network is not conducive to the detection of small ship targets. On the contrary, the shallow network contains more location information, but the semantic features are insufficient, which makes the shallow network not suitable for detecting large ships.

Inspired by Lin et al. [34], we propose BDFPN to extract the features of dense multiscale objects from SAR images as shown in Fig. 5. BDFPN includes bottom-up feed-forward networks, FFB, transverse connections and top-down up-sampling processes. BDFPN fuses the spatial features extracted from the spatial domain and high-frequency and low-frequency textural information extracted by the Haar wavelet transform, which can better extract the multiscale features of the ship target in two domains.

In the whole bottom-up process, the convolution layer is composed of five stages, and the outputs of {Conv1–Conv5} stages are selected as the reference set of feature maps. In the process of top-down and horizontal linking, the top-down network is to do double up-sample operations to generate feature maps with higher resolution, more semantic and textural information. Taking C4 as an example, a 1×1 convolution layer is added to C4 to generate P4. Then, to double up-sample P5 and add to the output of the 1×1 convolution layer. Next, utilizing a 3×3 convolution layer to generate P4. Among them, the additional 3×3 convolution layer is to reduce the aliasing effect in the upsampling process and the double upsample is to make sure these layers have the same spatial size. Finally, the FFB is used to fuse the final feature image in the spatial and frequency domain as M4.

We propose FFB to fuse the spatial and frequency features, which can refine the 2-D features, and details of FFB as shown in Fig. 2. First, we introduce spatial-frequency features through channel stacking. Second, by utilizing the residual-block-like module, we can retain more original features. Third, the attention module weights the feature maps of different scales to highlight the important features of specific scales, so that the two branches can complement each other effectively as shown in Fig. 6(a). Specifically, given the intermediate features S_i and F_i of the ith spatial block and frequency block, we use FFB to aggregate the feature graph into

$$M_i = \mathrm{FFB}(S_i \| F_i), \quad 1 \le i \le N \tag{5}$$

where $M_i \in 2C \times H \times W$ denotes the fused features, and $\|$ represents channel connection, we stack two features rather than element-wise addition for the input. Next, we build our FFB with 1×1 convolutional layer to focus on the channel-wise fusion. After the concatenate features get the salient features through the attention module, it is added to the extracted deep residual features with a conv adjusting

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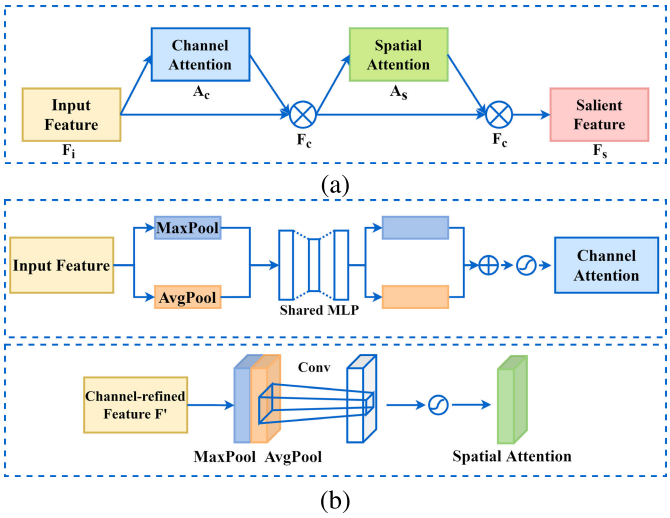


Fig. 6. (a) Structure of attention module, which mainly consists of channel attention and spatial attention. (b) Details of channel attention and spatial attention. The feature map passes through the MaxPool layer, AvgPool layer, and the share MLP module. After the ReLU activation function, the feature map is added element by element, and the channel attention result is obtained through a sigmoid activation function. The output results of channel attention are pooled to two feature maps through maximum pooling and average pooling, and the feature map of spatial attention is obtained through convolution and a sigmoid activation function.

the channel from $2C$ to C. Finally, generating the spatial-frequency fusion feature M_i, which is transferred to the final module subsequently.

For feature maps at the same scales, channel attention focuses on selecting feature maps of suitable scales, and spatial attention concentrates on salient regions containing targets in [38]. Therefore, global unblurred features are merged with local salient features to optimize fused feature maps for multiscale targets detection in different scenes, which is the main purpose for putting the attention module in the process of feature extraction.

It obtains attention maps along channel and spatial separately, as shown in Fig. 6(a), the input feature map F_i is multiplied by the channel attention map A_c and the spatial attention map A_s to obtain a salient feature map F_s. The overall process of attention module can be described as follows:

$$\begin{aligned} F_c &= A_c(F_i) \otimes F_i \\ F_s &= A_s(F_c) \otimes F_c \end{aligned} \tag{6}$$

where $F_c \in C \times H \times W$ is the feature map obtained by channel attention and $\otimes$ represents the element-wise addition.

The details of attention module showns in Fig. 6(b). The channel should pay attention to passing the input feature map through two parallel MaxPool layers and AvgPool layers, changing the size of the feature map from $C \times H \times W$ to $C \times 1 \times 1$, and then reducing the parameters through the share MLP module. Channel attention focuses on keeping channel dimensions unchanged and compressing spatial dimensions. It focuses on meaningful information in the input image.

The spatial attention module focuses on the specific location of salient features. The output results are pooled to two $1 \times H \times W$ feature maps through maximum pooling and average pooling, and then the two feature maps are concatenated. The 7×7 convn becomes the feature map of one channel, and then a sigmoid is used to get the feature map of spatial attention. Finally, the output results are multiplied by the original map and changed back to the size of $C \times H \times W$.

Therefore, the combination of these two attention modules will extract important information of the feature map of the detected ship, which can effectively eliminate false alarms in various scenes of SAR images, such as ground objects in inshore scenes. Then we exploit the RPN to generate a series of candidate regions with spatial-frequency features. And the location information of candidate region can be mapped into the feature maps and original image to obtain the corresponding ship targets image.

D. Loss Function

Multitask loss is applied to optimize RPN and final prediction network [43], which is the same as other classical two-stage object detection networks, as shown in the following equation:

$$L(p_i, t_i) = \frac{1}{N_{\text{cls}}} \sum_i L_{\text{cls}}\left(p_i, p_i^*\right) + \lambda \frac{1}{N_{\text{reg}}} \sum_i p_i^* L_{\text{reg}}\left(t_i, t_i^*\right) \tag{7}$$

where λ is a weighting parameter. In addition, p_i and p_i^* are predicted probability and ground-truth labels, respectively. Similarly, t_i and t_i^* are the predicted bounding box and ground-truth box, separately. N_{cls} and N_{reg} are the numbers of mini-batch in the training process, the classification loss and the regression loss are represented by $L_{\text{cls}}(p_i, p_i^*)$ and $L_{\text{reg}}(t_i, t_i^*)$. Cross entropy (CE) loss is utilized as the classification loss, the definition is as follows:

$$L_{\text{cls}}\left(p_i, p_i^*\right) = -\log\left(p_i^* p_i + \left(1 - p_i^*\right)(1 - p_i)\right). \tag{8}$$

The regression loss function is defined by

$$L_{\text{reg}}\left(t_i, t_i^*\right) = \text{smooth}_{L1}\left(t_i - t_i^*\right) \tag{9}$$

where

$$\text{smooth}_{L1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases}$$

IV. Experimental Results and Evaluation

In this section, the experiments are implemented to evaluate the detection performance of the proposed method. First, we introduce the dataset and the settings of related experiments. Next, the evaluation criteria are described. At last, we compare our method with other CNN-based methods using offshore and inshore ships to imply the superiority of the proposed method.

A. Dataset and Settings

SSDD is acquired through RadarSat-2, TerraSAR-X, and Sentinel-1 satellites [6], these data have four polarization modes, including HH, HV, VV, and VH. SSDD totally has 1160 SAR images with a resolution range from 1 to 15 m, containing multiscale ships in various environments

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TABLE I
DETECTION RESULTS OF DIFFERENT CNN-BASED METHODS ON SHIPS IN DIFFERENT SCENES

Method	Backbone	mAP	Inshore mAP	Offshore mAP
SSD [19]	ResNet50	83.34%	74.60%	96.27%
FPN [34]	ResNet50	86.17%	80.78%	93.92%
DAPN [33]	ResNet101	87.91%	81.29%	97.36%
ARPN [36]	ResNet101	89.34%	85.13%	95.24%
PolFourier [37]	ResNet50	94.08%	87.60%	98.25%
FTPN [45]	ResNet101	90.08%	84.41%	91.25%
Ours	ResNet50	**96.43%**	**91.10%**	**98.96%**

and different scenes (including 928 offshore scene images and 232 inshore scene images). There are 2456 ship targets in these images, and each image contains 2.12 ship targets on average. In our experiment, we divide SSDD into three parts (training, validation, and test set) with the proportion of 7:2:1 according to the official recommendations.

In this article, we use ResNet50 [27] pretrained on ImageNet dataset as the backbone network, the weight decay and the momentum are 0.0001 and 0.9, respectively. Besides, the learning rate is 0.001 with a decay ratio of 0.1. In addition, to ensure the reliability of the experiment, we set the maximum number of iterations for training to 20 000. All experiments in this article are completed in Pytorch. The experimental platform is configured with GPU Tesla V100.

B. Evaluation Metric

To evaluate the detection performance quantitatively, We adopt the standard metric COCO [44] to evaluate the performance of different ship detection methods on SSDD, including mean average precision (mAP) and mAP in inshore and offshore scenes, mAP is defined as

$$\text{mAP} = \int_0^1 P(R)dR \tag{10}$$

where $P(R)$ is the curve of precision–recall, they are obtained by four well-established components in information retrieval, true positive (TP), true negative (TN), false positive (FP), and false negative (FN). Moreover, the target detected by a bounding box is a correct ship, when the IoU between this bounding box and a single ground truth is higher than 0.5.

C. Experimental Results

In this section, we divide SSDD into two parts according to the inshore and offshore scenes, and use our algorithm in the two scenes to verify its effectiveness and high accuracy in the inshore scenes. At the same time, we compare the proposed method with other CNN-based methods.

Table I shows the detection results of different CNN-based methods tested on ships in different scenes, e.g., inshore, offshore and mixed scene. The first two rows of Table I shows The detection results of classical CNN-based methods, e.g., SSD [18] and FPN [33]. The third to the fifth rows of Table I are the detection results of CNN-based methods specifically designed for object detection in SAR images, e.g., DAPN [32], ARPN [35] PolFourier [36], and FTPN [45].

In Table I, FPN performs better than SSD in offshore scene. As a one-stage detector, SSD directly predicts locations and class labels of the ship targets at several feature maps without RoI proposals, which has a simpler network structure than Faster RCNN. So its performance of detecting ships may be more stable in simple offshore background. The performance of our method is better than that of FPN by a large margin, it might be because of the contributions of feature pyramids adopted and the weak feature extraction abilities of Faster-RCNN. Furthermore, in terms of offshore mAP, compared with the CNN-based methods designed for object detection in SAR images, the score of our method is 1.6%, 3.72%, 0.71%, and 7.71% higher than that of DAPN, ARPN, PolFourier, and FTPN, respectively.

The detection results of different CNN-based methods tested on inshore ships are shown in the fourth column in Table I. Due to much interference around inshore ships, the accuracy of SSD decreases obviously. Although feature pyramids are adopted in FPN, DAPN, ARPN, and FTPN for extracting multiscale features, the inshore mAP of our method is still 10.32%, 9.81%, 5.97%, and 6.69% higher than the scores of these methods. Besides, the mAP of our method is the highest compared with the methods designed for object detection in SAR images. It might be because of the well-designed feature extraction and spatial-frequency feature fusion in our method.

Fig. 7 shows the detection results of different methods tested on offshore and inshore ships, respectively. At the first condition in offshore scene (the first row of Fig. 7), which contains multiple ship targets per image. SSD misses two ship targets, probably because its simple structure is not suitable for detecting small-scale ship targets. Other algorithms have correctly detected all targets, except for the ship detection result cofidence of Fig. 7(e) are 60% and 74%, this may be because the two-stage detector is more accurate than the one-stage detector, which uses an appropriate feature pyramid network. Therefore, we can obtain abundant information on small-scale ships on low-level feature maps. In the second condition in offshore scene (the fifth row of Fig. 7), ships and surroundings are similar, e.g., isles and lands. Isles are detected as ships by DAPN, and one ship is detected as several ships by DPAN and ARPN. This may be due to the weak ability to distinguish ships and interference. However, our method could detect these ships without making any mistakes, which means our method has strong and robust feature representation abilities. In the last condition in offshore scene (the sixth row of Fig. 7), there are ship wakes and lots of noise. DAPN does not detect the ship, Haar wavelet transform helps our method represent and discriminate ships and wrong distinctly to detect these ships correctly without missing.

Besides, three groups of detection results tested in different conditions of inshore ships are shown in Fig. 7. In the first condition (the second row of Fig. 7), some large-scale inshore ships. Most isles are detected as ships, although feature pyramid networks are used in FPN, DAPN, and ARPN to obtain multiscale refined features, the single spatial feature in the feature map has a limited impact on the final detection results.

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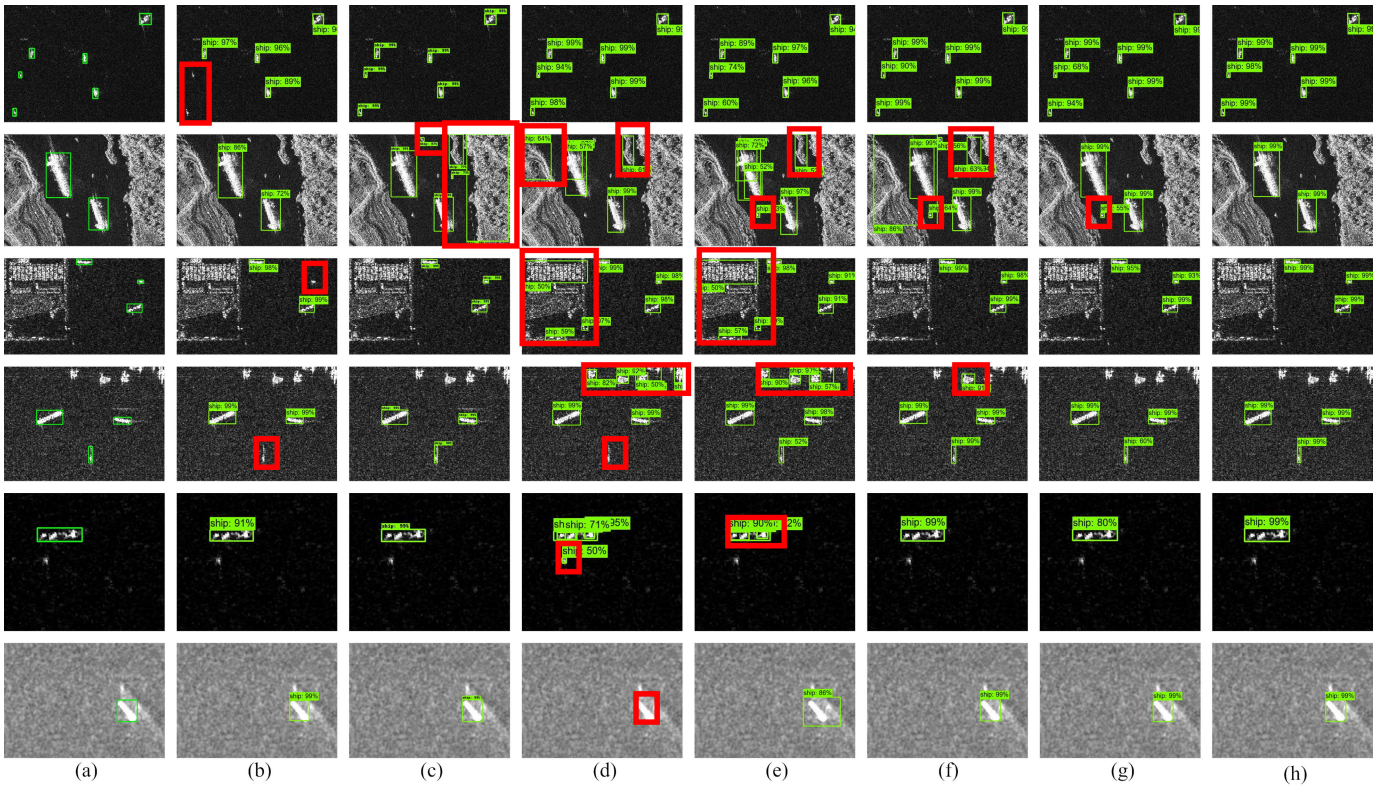


Fig. 7. Detection results of different methods tested on offshore and inshore ships, respectively. (a) Images include ground truth ships. (b)–(h) Detection results on offshore ships of SSD, FPN, DAPN, ARPN, PolFourier, FTPN, and the proposed method, respectively. Rectangles with red color are error predictions and undetected ship targets. In the inshore scene, our algorithm has higher accuracy than other methods.

In the second condition (the third row Fig. 7), multiscale ships exist in complex surroundings. A small ship is missed by SSD, while DAPN and ARPN treat objects on the coast as ships, possibly because they are similar to ships. In the last condition (the fourth row of Fig. 7), isles around ships. And a small ship with a wake is missed by SSD. DAPN not only misses a ship, but also regards some islands as ship targets. The locations of ships predicted by ARPN and PolFourier are inaccurate, compared with the ground-truth ships. Our method enriches spatial features through contour and detail features, which can detect these multiscale ships with high probability and positioning accuracy.

Overall, making use of more abundant feature information, which contains spatial features and frequency features without noise and retains details and contours, our algorithm can clearly and accurately point out the specific position of the ships, no matter in offshore scene or in inshore scene.

V. Discussion

In this section, we discuss the advantages and disadvantages of our algorithm in ship detection in inshore scene. And we set up an ablation experiment to verify the effectiveness of the fusion strategy, comparing detection performances of spatial features, frequency features, and fused features.

A. Comparison in Inshore Scene

To more intuitively see the performance of our algorithm on a single image, we make statistics on the detection results of the algorithm in inshore scene (including 232 images). The ratio of the number of errors to the total number is shown in Table II, Figs. 8 and 9 show the advantages and some problems of our algorithm in inshore scene images.

TABLE II

Statistics of Undetected, False, and Undetected and False Condition in All Detection Results

Method	Undetected	False	Undetected&False
SSD [19]	46.05%	**1.75%**	7.02%
FPN [34]	19.74%	32.02%	11.84%
DAPN [33]	10.53%	50.00%	21.93%
ARPN [36]	12.28%	46.05%	22.37%
PolFourier [37]	**3.51%**	55.26%	**3.51%**
FTPN [45]	20.61%	8.77%	6.58%
Ours	7.89%	21.49%	7.89%

In Table II, Undetected means that only some ships are not detected in one image. False means that only other objects are recognized as ships in one image. Undetected&False means that both undetected and false occur in one image. We can see that although the undetected rate is not as low as that of PolFourier, the false detection rate is 19.74%, which is higher than that of SSD, and the score of undetected & false detection, which ranks third among all algorithms is 7.89%. Overall, our algorithm has a good performance.

As shown in Fig. 8, ships are closely distributed in the inshore scene. Although SSD can detect some ships, it has

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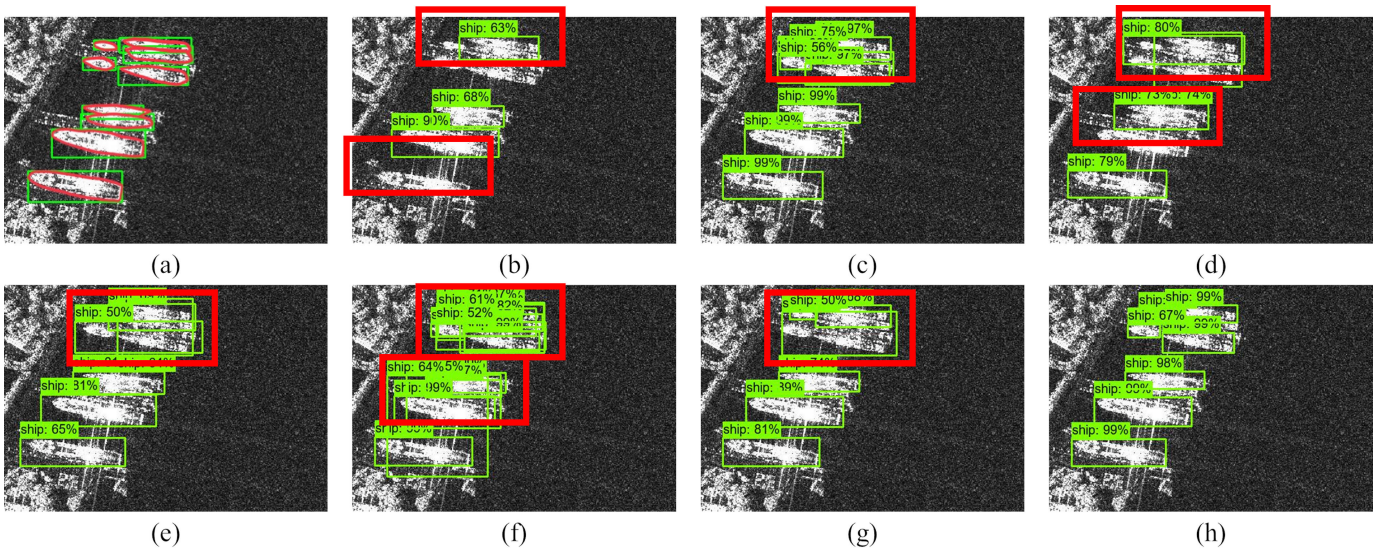


Fig. 8. Ships are closely distributed in the offshore scene. (a) and (e) Ground truth. (b), (c), (d), (f), (g), and (h) Detection results on offshore ships of SSD, FPN, DAPN, ARPN, PolFourier, FTPN, and the proposed method, respectively. Rectangles with red color are error predictions and undetected ship targets.

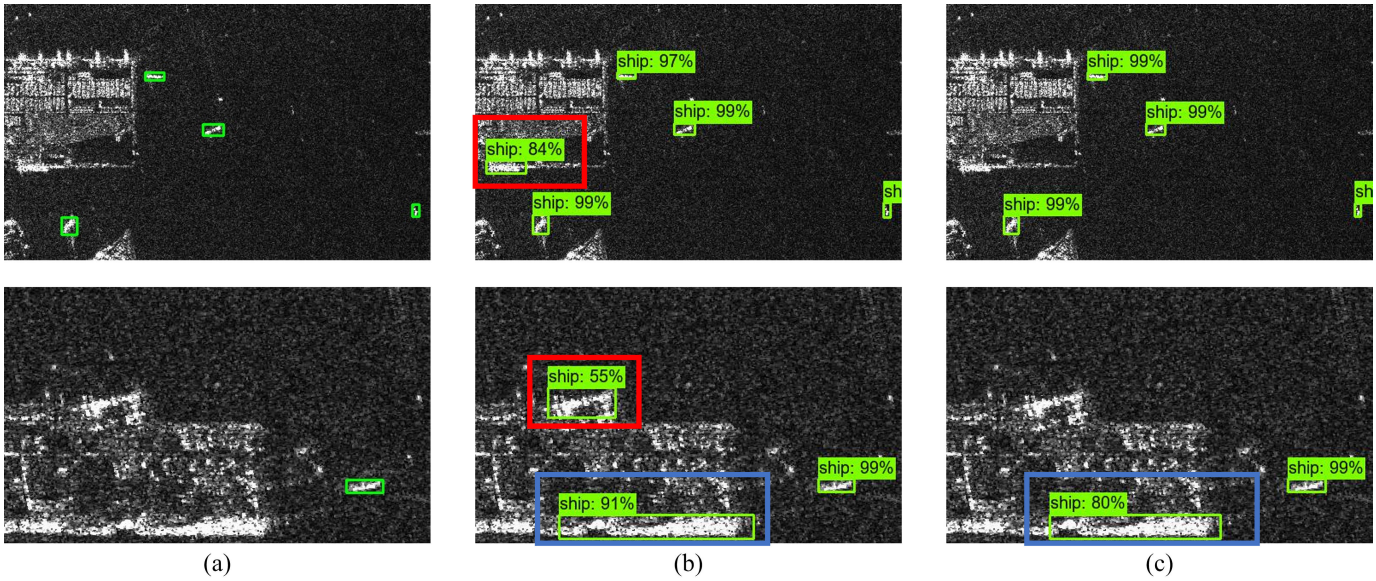


Fig. 9. Comparison of detection results between PolFourier and our method in similar scenes, as well as the probability distribution histogram of real targets and false detected objects. (a) Ground truth. (b) and (c) Detection results on inshore ships of PolFourier and the proposed method, respectively. The red box indicates other objects are detected as ships by PolFourier, and the blue box indicates that both PolFourier and our algorithm detect other objects as ships.

failed to detect multiple ships and cannot distinguish two ships that are very close. FPN and ARPN did not miss any ships, but the rectangle in red was recognized as a whole ship. DAPN, PolFourier and FTPN have the same problem as FPN and ARPN. Meanwhile they also undetected some ships. We can clearly distinguish ship targets from the neighbor objects whose textures are extremely similar to ships, especially in inshore scene, using refine feature pyramid network and spatial-frequency fusion feature to provide more information.

The comparison of detection results between PolFourier and our method in similar scenes, as shown in Fig. 9. Although PolFourier solves the rotation problem of ship targets through polar Fourier transform, for frequency domain features, the highlighted part in the image will become more obvious, like the coast in the red rectangle, which is regarded as a ship by PolFourier, and the confidence of the true detected result is higher than 80%. Haar wavelet transformation makes up for the lack of spatial features in ship contour and details with the high-low-frequency information extraction. Our algorithm still gives FPs for inshore objects that are very similar to ships from skewness coefficient to object shape. Although our algorithm occasionally detects the coast as a ship, our confidence is much lower than that of the PolFourier, as shown in the blue rectangle.

B. Ablation Experiment

In this section, the performance of binary domain features is evaluated, where FPN is used as the baseline. We also

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TABLE III
DETECTION RESULTS OF RSF, RFF, AND RBDF ON SHIPS

Feature Type	Attention	mAP
RSF		86.17%
RSF	✓	**89.52%**
RFF		87.40%
RFF	✓	**91.79%**
RBDF		94.21%
RBDF	✓	**96.43%**

verify the performance of the proposed FFB. Table III shows the mAP of ResNet with spatial features (RSF), ResNet with frequency features (RFF), and ResNet with binary domain features (RBDF) on ships.

According to Table III, the mAP of RSF, RFF, and RBDF with attention module are 3.35%, 4.39% and 2.22% higher than RSF, RFF, and RBDF without attention module, respectively. This also verifies that no matter which type of feature is used, the attention module improves the representation capability of concatenate feature maps and can detect multiscale ships more effectively. The mAP of RBDF with attention module is 6.91% and 4.64% higher than that of RSF and RFF in Table III, respectively. As the information is obtained by spatial features and frequency features is relatively simple and unique. There is omission or redundancy in the information acquisition of objects with textures similar to ships, which makes some error alarms are not suppressed effectively. Compared with offshore ships, complex environments such as docks, buildings and vehicles may cause a great effect on the accurate detection of inshore ships. Thus, the detection performance is interfered. Since the binary domain features are the result of complementary and perfect information in the spatial domain and frequency domain, which makes it can effectively reduce most false targets with similar textures when detecting ships. For this reason, the detection result of binary domain features is much more accurate than a single feature.

VI. CONCLUSION

In this article, we propose a two-stage detector called MFFN to relieve the background interference for detecting multiscale and inshore ships in SAR images. We design a BDFPN with FFB to obtain the multiscale features of the ship target from a top-down pyramid structure and suppress interference of surroundings, and fuse the semantic features extracted from the spatial domain with the textural and detail features extracted from the frequency domain. In the experiment, we prove superiorities of our method by evaluating the performance of our method with some CNN-based methods on the SSDD. Besides, we further compared our method with other SAR ship detections in inshore scene, which proves our method could detect multiscale ships with preferable performance and competitive generalization.

Our algorithm is not easily affected by sea waves and complex background, and can accurately distinguish ship wakes, other objects and ship targets in inshore scene, and ships near the coast are not easy to be ignored. The performance of SAR ship detection is not only unrestricted, but also suitable for detecting multiscale ships in inshore scene.

In the future, we will focus on combining the rotation invariance of ship targets in SAR images with an anchor-free convolution network, and even perform morphological preprocessing to further improve the detection accuracy in inshore scenes.

REFERENCES

[1] N. Wang, B. Li, X. Wei, Y. Wang, and H. Yan, "Ship detection in spaceborne infrared image based on lightweight CNN and multisource feature cascade decision," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4324–4339, May 2021.

[2] X. Leng, K. Ji, S. Zhou, and X. Xing, "Ship detection based on complex signal kurtosis in single-channel SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6447–6461, Sep. 2019.

[3] S. Brusch, S. Lehner, T. Fritz, M. Soccorsi, A. Soloviev, and B. van Schie, "Ship surveillance with TerraSAR-X," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 1092–1103, Mar. 2011.

[4] A. Renga, M. D. Graziano, and A. Moccia, "Segmentation of marine SAR images by sublook analysis and application to sea traffic monitoring," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1463–1477, Mar. 2019.

[5] T. Zhang and X. Zhang, "ShipDeNet-20: An only 20 convolution layers and <1-MB lightweight SAR ship detector," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 7, pp. 1234–1238, Jul. 2021.

[6] J. Li, C. Qu, and J. Shao, "Ship detection in SAR images based on an improved faster R-CNN," in *Proc. SAR Big Data Era, Models, Methods Appl. (BIGSARDATA)*, Nov. 2017, pp. 1–6.

[7] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–216, Jan. 1992.

[8] A. C. Frery, H.-J. Müller, C. C. F. Yanasse, and S. J. S. Sant'Anna, "A model for extremely heterogeneous clutter," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 3, pp. 648–659, May 1997.

[9] C. P. Schwegmann, W. Kleynhans, and B. P. Salmon, "Manifold adaptation for constant false alarm rate ship detection in South African oceans," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 7, pp. 3329–3337, Jul. 2015.

[10] X. Qin, S. Zhou, H. Zou, and G. Gao, "A CFAR detection algorithm for generalized gamma distributed background in high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 4, pp. 806–810, Jul. 2013.

[11] C. Wang, Z. Wang, H. Zhang, B. Zhang, and F. Wu, "A PolSAR ship detector based on a multi-polarimetric-feature combination using visual attention," *Int. J. Remote Sens.*, vol. 35, no. 22, pp. 7763–7774, 2014.

[12] G. E. Atteia and M. J. Collins, "On the use of compact polarimetry SAR for ship detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 1–9, Jun. 2013.

[13] X. Wang and C. Chen, "Ship detection for complex background SAR images based on a multiscale variance weighted image entropy method," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 2, pp. 184–187, Feb. 2017.

[14] Z. Zou and Z. Shi, "Ship detection in spaceborne optical image with SVD networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5832–5845, Oct. 2016.

[15] G. Ferrara, M. Migliaccio, F. Nunziata, and A. Sorrentino, "GK-based observation of metallic targets at sea in full-resolution SAR data: A multipolarization study," *IEEE J. Ocean. Eng.*, vol. 36, no. 2, pp. 195–204, Apr. 2011.

[16] C. Wang, F. Bi, L. Chen, and J. Chen, "A novel threshold template algorithm for ship detection in high-resolution SAR images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 100–103.

[17] J. Zhu, X. Qiu, Z. Pan, Y. Zhang, and B. Lei, "Projection shape template-based ship target recognition in TerraSAR-X images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 2, pp. 222–226, Feb. 2017.

[18] M. Zhu, G. Hu, H. Zhou, and S. Wang, "Multiscale ship detection method in SAR images based on information compensation and feature enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5117913.

[19] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. 14th Eur. Conf. Comput. Vis.—ECCV*. Amsterdam, The Netherlands: Springer, Oct. 2016, pp. 21–37.

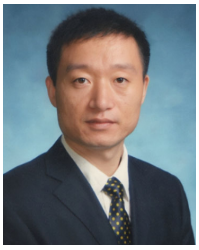
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[20] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.
[21] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
[22] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020.
[23] Y. Mao, Y. Yang, Z. Ma, M. Li, H. Su, and J. Zhang, "Efficient low-cost ship detection for SAR imagery based on simplified U-Net," *IEEE Access*, vol. 8, pp. 69742–69753, 2020.
[24] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.—MICCAI*. Munich, Germany: Springer, Oct. 2015, pp. 234–241.
[25] L. Yang, J. Su, H. Huang, and X. Li, "SAR ship detection based on convolutional neural network with deep multiscale feature fusion," *Acta Opt. Sinica*, vol. 40, no. 2, 2020, Art. no. 0215002.
[26] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[27] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[28] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1–9.
[29] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.
[30] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
[31] Y. Gui, X. Li, and L. Xue, "A multilayer fusion light-head detector for SAR ship detection," *Sensors*, vol. 19, no. 5, p. 1124, Mar. 2019.
[32] Z. Lin, K. Ji, X. Leng, and G. Kuang, "Squeeze and excitation rank faster R-CNN for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 5, pp. 751–755, May 2018.
[33] Z. Cui, Q. Li, Z. Cao, and N. Liu, "Dense attention pyramid networks for multi-scale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8983–8997, Nov. 2019.
[34] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.
[35] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.
[36] Y. Zhao, L. Zhao, B. Xiong, and G. Kuang, "Attention receptive pyramid network for ship detection in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2738–2756, 2020.
[37] D. Li, Q. Liang, H. Liu, Q. Liu, H. Liu, and G. Liao, "A novel multidimensional domain deep learning network for SAR ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5203213.
[38] D. Shi, L. Zheng, and J. Liu, "Advanced Hough transform using a multilayer fractional Fourier method," *IEEE Trans. Image Process.*, vol. 19, no. 6, pp. 1558–1566, Jun. 2010.
[39] J. Dai, K. He, and J. Sun, "Instance-aware semantic segmentation via multi-task network cascades," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3150–3158.
[40] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," *J. Mach. Learn. Res.*, vol. 15, no. 4, pp. 315–323, 2011.
[41] R. C. Gonzales and P. Wintz, *Digital Image Processing*. Reading, MA, USA: Addison-Wesley, 1987.
[42] S. G. Mallat, "A theory for multiresolution signal decomposition: The wavelet representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 11, no. 7, pp. 674–693, Jul. 1989.
[43] T. Zhang, X. Zhang, J. Shi, and S. Wei, "HyperLi-Net: A hyper-light deep learning network for high-accurate and high-speed ship detection from synthetic aperture radar imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 123–153, Sep. 2020.
[44] T. Y. Lin, M. Maire, S. Belongie, J. Hays, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
[45] Z. Zhou, Z. Cui, Z. Cao, and J. Yang, "Feature-transferable pyramid network for dense multi-scale object detection in SAR images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 647–650.



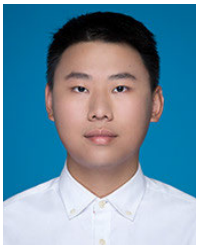
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